

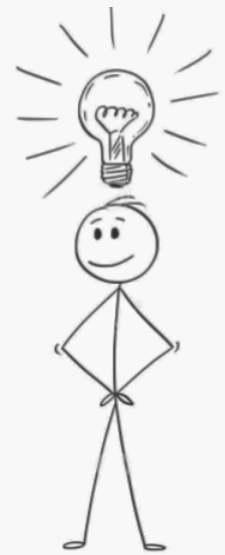
Three high-profile cases and one next-generation approach to integrating impact into financial analysis

While techniques for measuring social and environmental outcomes continue to improve, investors have lacked a framework for quantifying their impact in a way that can be coherently compared to financial returns. This has led to silos that optimize for one narrow outcome or another, rather than the combined financial and impact performance.

Impact Returns are a solution to this problem. They empower investors to make decisions that optimize total impact-adjusted returns—financial plus impact—leading to the best possible long-term, combined value.



How can I integrate impact into my investment process in a way that is genuine, not greenwashing? What about additionality? What about indirect effects?...



Impact
Returns

This report illustrates the assessment of *Impact Returns* for three cases: two classic examples and one stylized example based on a recent investment opportunity. The goals of the report are:

1. To provide concrete examples of impact returns.
2. To demonstrate that impact returns can be applied across asset classes and areas of impact.
3. To serve as a guide for assessing the impact returns of new opportunities.

One way of understanding impact returns is as the answer to the question: "How much additional financial return would an otherwise identical investment, without investor impact, need to offer to be the better investment?". This means that with valid impact returns you can say things like:

"This investment has a 15% financial return plus a 5% impact return, for an impact-adjusted return of 20%. Given my goals, this is better than an alternative investment with a 19% return (e.g. 19% financial return + 0% impact return) for the same risk."

For the cases in this report we don't present financial return estimates, but rather leave it to the reader to decide how the impact returns in each case might affect:

- **Trade-offs.** If the risk-adjusted financial return offered by an opportunity is below 'market rate', is the impact return sufficient to make the investment worth doing?
- **Investment sizing.** If there are no upfront trade-offs, does the impact return justify making a bigger, more concentrated investment than if impact wasn't a consideration?

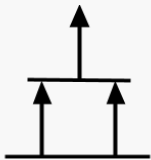


Building on the work of pioneers in the financial, impact and academic worlds, we've identified 'Three Keys' to getting valid impact returns that can be combined with financial returns.



Key 1 – Magnitude: Assess the magnitude of all material outcomes.

- Indeed, for these cases, there is a wide variation in the magnitude of the outcomes.



Key 2 – Contribution: Adjust for the 'contribution' of each actor in the impact pathway, relative to what would have otherwise happened, in a way that represents positive-sum, non-binary thinking.

- For these cases, this means assessing '*enterprise impact*', the difference between the expected outcomes with and without the investee, and '*investor impact*', the difference between enterprise impact with and without the investment.
- We have developed a formula to do this based on scoring an opportunity on what we call its '*crowdedness*' and '*scalability*'. Crucially, this formula is designed to address common concerns about assessing contribution. It is about positive-sum thinking (not zero-sum attribution) and non-binary (e.g. not just 0% or 100%).
- In these cases we focus on financial contribution. Non-financial contributions are also important, but we don't have sufficient context to reasonably assess this for these cases.



Key 3 – Strategic Pricing: Translate investor impact into a financial value using a 'strategic price of impact' (SPI) that is informed by opportunity costs not social costs.

- The SPI is not a normal price, but a 'strategic' one just for assessing impact returns and making decisions. It is different from and complementary to approaches like social returns and cost-benefit analysis.
- In this report, this takes the form of a strategic price of impact based on the estimated impact per dollar of GiveWell top charities. These charities are well-known impact benchmarks.

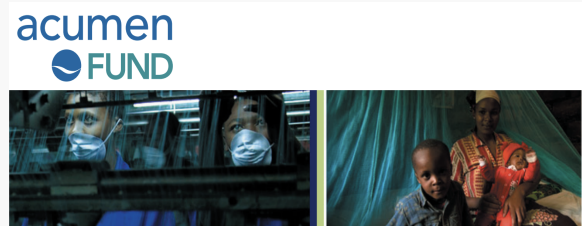
Our team has developed an open-source methodology for assessing impact returns that successfully puts these 'Three Keys' into practice. We've iteratively refined this methodology by applying it to real investments. We're releasing this case study report to share this methodology and demonstrate it using well-known cases. This report begins with descriptions of the cases, followed by descriptions of each step in the impact return assessment process, the resulting impact returns and a discussion.

Finally, we encourage you to create your own version of these analyses using [this template](#).

The Cases

Opportunity 1: Mosquito Nets

A loan by Acumen to A to Z Textile Mills in Tanzania in 2003. \$455k (including operational costs) was provided to finance the local manufacturing of modern long-lasting insecticide-treated bednets (LLINs) that are effective against the mosquitos that transmit malaria. The original case study is [here](#).



Opportunity 2: Education Technology



This case is about TPG Rise's 2017 private equity investment in EverFi, an online education platform with programs that seek to have positive impacts on alcohol abuse, sexual assault and financial literacy. For simplicity our model focuses on the AlcoholEdu program, a course designed to deter alcohol abuse among college students. We then estimate EverFi's overall impact by assuming AlcoholEdu is the same portion of EverFi's impact as calculated by TPG Rise. The original case study is [here](#).

Opportunity 3: Covid Vaccine

A hypothetical investment in 2021 in a venture that aims to rapidly develop a low-cost and easily-distributed vaccine for future variants of Covid-19. The success of this venture would result in i) producing a vaccine that works for the billions of people in low-income countries, and ii) proving that such a vaccine could be rapidly developed, paving the way for future vaccine developments to achieve this pace.



Impact Returns are a Middle Way

Impact Returns are a 'middle way'. They avoid the extremes of common views that don't result in impact values that can be coherently combined with financial values. The table below presents how this is true for the three cases in this report.

The table highlights how we have built on the methodologies in these pioneering cases, keeping the ideas from each case that are in line with the concept of impact returns. We then added the concept of 'contribution' and applied each concept consistently across each case.

One Extreme	 Impact Returns	Other Extreme
Many people seem to act like impact shouldn't be measured, analyzed or projected	Key 1 – Magnitude: As TPG Rise originally did for opportunity 2, for each case we project the magnitude of outcomes out to a 10 year horizon	Acumen originally only considered impact during the term of the loan
Many philanthropists write off cases like these without really understanding them, as they don't believe it's possible for return-generating opportunities to have 'additionality'	Key 2 – Contribution: For these cases, we apply 'contribution multipliers' from 9% to 71%, for both enterprise and investor contribution	Acumen and TPG Rise originally assumed 100% 'contribution multipliers' for opportunities 1 & 2
Many investors would price impact at \$0, either because they don't value it or because they don't want to integrate impact with financial returns	Key 3 – Strategic Pricing: As Acumen originally did for opportunity 1, we price impact based on the cost to save lives by donating malaria nets	TPG Rise originally presented the impact value of opportunity 2 in terms of the social cost

Overview of the Impact Return Assessment Process

The diagram below outlines the steps involved in the assessment of impact returns. The steps can be rapidly completed in a back-of-the-envelope way, or with deep, careful analysis.

1	Potential Gross Impact	<u>IMP Dimensions</u> What Who How Much	Estimate the total potential future impact of the underlying enterprise in different scenarios.
	Expected Gross Impact	How Much Risk	Apply probabilities to the Potential Gross Impact scenarios.
	Enterprise Effectiveness		Divide Expected Gross Impact by the financial 'size' of the underlying enterprise.
2	Contribution Multipliers	Contribution	Adjust for the contribution of each actor in the impact pathway, especially investor contribution.
3	Strategic Pricing		Convert to returns based on opportunity costs and the expected investment period.

Potential Gross Impact is the 'absolute' impact of an opportunity. That is, without adjustments for contributions or risk. It is the kind of result that you see most enterprises report (e.g. as ESG metrics). This relates to the *Who*, *What* and *How Much* [dimensions of impact](#).

Expected Gross Impact is assessed by applying explicit probabilities to *Potential Gross Impact*. We recognize that a lot of people don't have training in forming and using explicit, subjective probabilities (though it is very common, for example, in the world of hedge funds and proprietary trading). However, we see it as best practice to develop the ability to use probabilistic reasoning. In particular it is transparent - much more so than if it was simply reported that an impact is "pretty likely" (what would that really mean?).

Expected Gross Impact relates to the *How Much* and *Risk* dimensions of impact. Some investors see *Risk* as only reflecting the probability that the potential impacts are not realized. '*Risk*' can also mean a preference for impacts that are less volatile - this can be reflected by applying lower 'risk-adjusted' probabilities for more uncertain potential impacts.

Enterprise Gross Impact per dollar. To prepare for forming a 'return' in the final step we want to understand how much impact is produced per dollar. So we divide the *expected gross impact* by the expected financial size of the enterprise. Different ways of assessing financial size have their own pros and cons. In this report we simply use estimates of the post-money equity valuation.

Contribution multipliers. Each actor in the impact pathway only makes a partial contribution to the expected gross impact. Without a given actor, another one may have stepped in to serve the same role. If we don't invest, then another investor may (partially) make up for this. In turn, if the enterprise we fund didn't exist, then another enterprise may have got funding to generate similar impacts.

The contribution multipliers associated with each actor multiply together along the impact pathway - in the cases here we multiply together the enterprise contribution and investor contribution multipliers. In general, there can be fund contribution multipliers or multipliers for other intermediaries. See the next subsection for more on contribution multipliers.

Note that, for simplicity, in this report we focus only on direct, financial investor contributions. In general, investors can also make important non-financial and indirect financial contributions.

Strategic pricing. The *Strategic Price of Impact* (SPI) represents the rate at which you are willing to trade financial returns for impact across your opportunities. It is obviously what you are doing if you seek to generate impact via charitable or other concessionary opportunities (including taking on above market risk). Even if you only invest in 'market-rate' opportunities, it is still what you are doing if you don't construct your portfolio to purely maximize (risk-adjusted) financial returns.

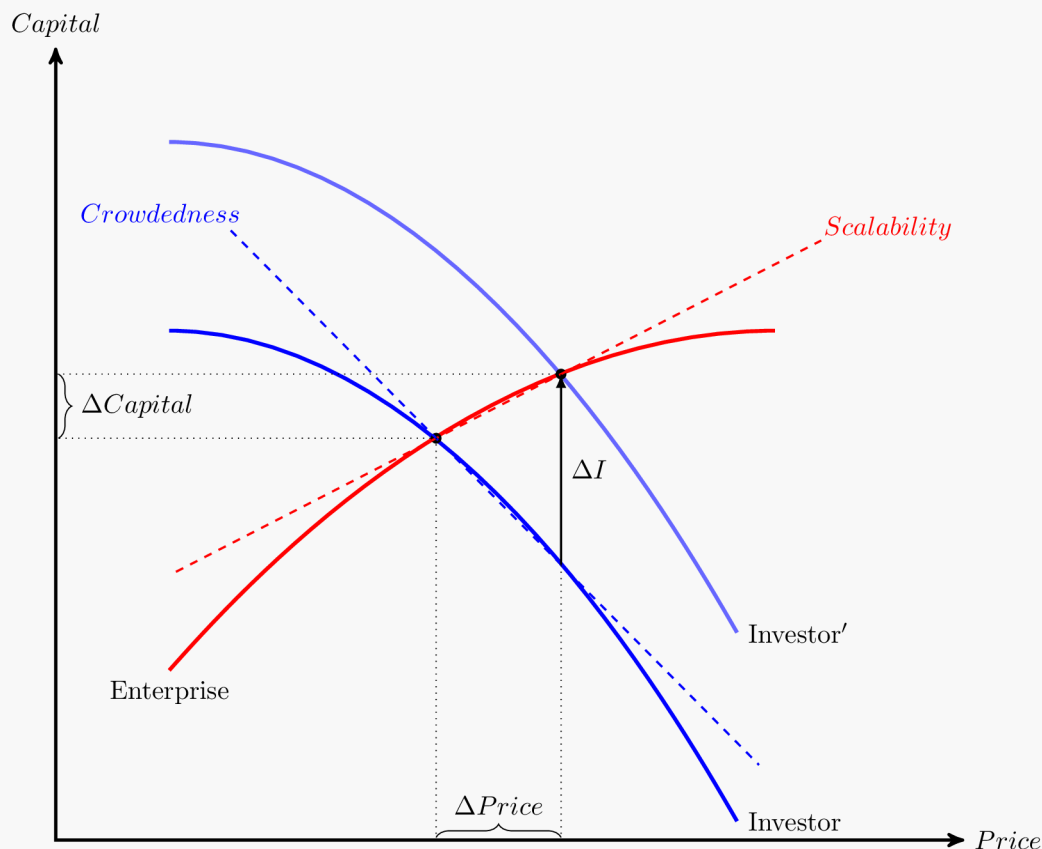
Translating impact into a financial value generates an **Impact Value**, which can be thought of as an 'impact cash flow' that occurs at the time the investment is made. The **Impact Return** represents the impact value as a percentage of the actual financial cost of the investment. Dividing by the expected holding period gives annualized impact returns – though these annual returns are a simplification. If you have full knowledge of all the expected cash flows or returns, both financial and impact, then incorporating them all into a single IRR or NPV calculation is the ultimate way to integrate impact into financial analysis.

The point of having an explicit SPI is to use it consistently across opportunities. It is 'strategic' because a) it will depend on your strategic context (e.g., what opportunities you have available, your impact goals), and b) it is only for the purposes of your decision making (you can't actually buy or sell impact). It should be informed by your opportunity costs and, thus, not be greater than the cost per unit impact of the best alternative investment you are willing to make. This means that, in general, the SPI should be lower than the social cost, because social costs are typically estimated based on the maximum society would be willing to pay for an outcome, not what we have the opportunity to pay.

Contribution Scores

We assess contribution in terms of two factors: **Scalability** and **Crowdedness**. *Scalability* is how productively an opportunity can use additional investment right now. *Crowdedness* is the wealth of the investors considering the opportunity relative to the perceived risk. Because of these two factors, it is not necessarily the least crowded opportunities that offer the best investor contribution.

The stylized diagram below illustrates how Scalability and Crowdedness are ways of interpreting a basic model of supply and demand. The **Investor curve** is the amount of units of production (*Capital*) the investment market is willing to buy given the *Price* at which the enterprise sells ownership in each unit. The **Enterprise curve** is the cost for the enterprise to produce new units of *Capital* as the total amount increases. The market clears at the point where these curves intersect and the amount of *Capital* determines how much the enterprise can produce and how much impact it will have.



Suppose you commit additional investment ΔI , thereby shifting the Investor curve to Investor'. This changes the point where the market clears, shifting *Price* by $\Delta Price$ and *Capital* by $\Delta Capital$. These shifts are driven by Scalability, which is the slope of the **Enterprise curve**, and Crowdedness, which is (minus) the slope of the **Investor curve**.

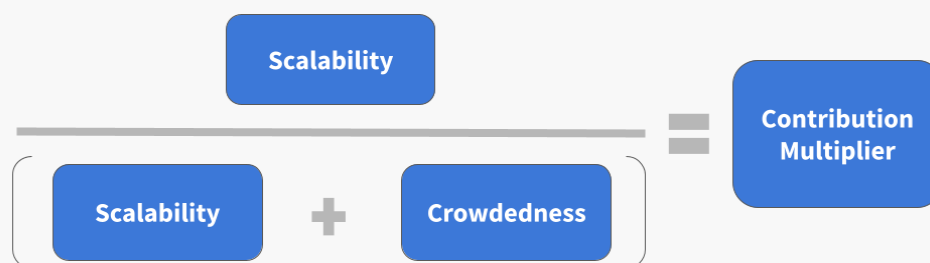
Basic geometry can be used to work out that the price change depends on both slopes:

- $\Delta Price = \Delta I / (Scalability + Crowdedness)$

In turn, because the market clearing point moves along the **Enterprise curve**:

- $\Delta Capital = \Delta Price \times Scalability = \Delta I \times Scalability / (Scalability + Crowdedness)$

The ratio of $\Delta Capital$ to ΔI is the 'contribution multiplier'. If Scalability increases, then the price change will decrease, but the capital change will increase and get closer to ΔI . But, if Crowdedness increases then both the price change and the capital change will decrease. The value of the contribution multiplier will be between 0% and 100%, meaning it avoids the 'binary', all-or-nothing nature of other definitions of 'additionality'.



Contribution NOT attribution: How contribution multipliers represent 'positive-sum' thinking

The discussion of the diagram above shows that the impact of the change ΔI is determined by the contribution multiplier. In our [technical paper](#) we also show that this is the optimal way for individual investors to approach their contribution, in terms of producing the optimal combination of impact and financial outcomes. Unlike attribution approaches, which give credit for impact like it's a zero-sum game, this doesn't say anything about the contribution of other investors or stakeholders. Instead, this approach represents 'positive-sum' thinking because it's about getting each investor to maximize their contribution, while respecting their unique financial goals and constraints. Double counting is not a concern because we're not applying traditional zero-sum logic.

It's about looking at the contributions of other investors as something to build on. And simply pointing out that this is harder when more investors are already making a contribution ('Crowdedness') and easier when there is more potential for additional contributions ('Scalability').

In general, *Scalability* and *Crowdedness* need to be assessed heuristically, though they can be informed by quantitative evidence. To make this easier, we assign S-scores and C-scores with 0 corresponding to the average, medium-sized public U.S. company. These scores are then converted into Scalability and Crowdedness values based on powers of ten¹. That is, an increment of 1 point in the S-score represents a 10x *increase* in scalability. Hence a S-score of 2 is 'highly scalable' (e.g. a business on the scale of Amazon) while S-scores down to -5 are reasonable for very early stage companies that can't absorb much capital right now.

To make it so that for both scores indicate higher contributions when they are more positive, a 1 point increment in the C-score represents a 10x decrease in Crowdedness. So, a C-score of 5 is an

¹ Scalability = 10^S and Crowdedness = 100×10^{-C} .

extremely neglected and uncrowded opportunity. C-scores that are negative represent a very crowded opportunity.

		Opportunity 1 (Mosquito Nets)	Opportunity 2 (Education technology)	Opportunity 3 (Covid Vaccine)
Enterprise Contribution	S-score	-2.5	1	-2
	C-score	3 to 4	0 to 1	4 to 5
	Multiplier	10%	24%	74%
Investor Contribution	S-score	-4 to -3	0	-3 to -2
	C-score	4 to 5	1 to 2	4 to 5
	Multiplier	11%	22%	47%

The positive relationship between contribution and financial returns

In the diagram above, Scalability and Crowdedness can be thought of as tilting the Enterprise and Investor curves around fixed points at the bottom of the diagram (i.e., where the curves start for low Capita values). These fixed points will be determined by the raw setup costs and profitability of the enterprise, before diminishing returns to scale start to bite. This means that opportunities with higher Crowdedness (steeper Investor curves) will tend to have higher prices. While opportunities with higher Scalability (steeper Enterprise curves) will tend to have lower prices.

This suggests a negative relationship between contribution multipliers and prices, and hence, a positive relationship between contribution multipliers and expected financial returns.

All else equal, the best opportunities will have high Scalability and low Crowdedness.

		Crowdedness	
		Lower	Higher
Scalability	Lower	Average contribution Average prices	Low contribution High prices
	Higher	High contribution Low prices	Average contribution Average prices

Enterprise Contribution. When assessing enterprise contribution, we note that Opportunity 2 (Education Technology) is the most scalable given its growing, largely online market that can deploy a lot of capital. Opportunity 1 and 3 both have negative scalability scores to represent the nature of their enterprises not being scalable in the short-term with LLIN demand uncertain and a new, unproven entrant in the Covid Vaccine space lacking immediate scalability. Opportunity 1 and 3, however, are in much less crowded industries being earlier stage, private, and focused on emerging markets.

Engagement a silver bullet?

We've noticed a trend in recent years of many investors assuming their financial contribution is very low, so they propose to focus on engagement and other non-financial activities instead. However, if everyone does this then that makes engagement a rather crowded activity! We recommend that investors assess the Scalability and Crowdedness of all investments they make, whether money as a financial contribution or time and expertise as a non-financial contribution.

Investor Contribution. Both Opportunities 1 and 3 are assigned low scalability and low crowdedness scores to reflect their inherently weaker ability to deploy more capital in the near term. On the other hand, both are relatively neglected, with only a limited pool of investors being willing and able to make such investments. Even if these investors have billions to invest, these opportunities are competing for these billions, whereas public firms are competing for a share of trillions of dollars of capital. Opportunity 2 is assessed to have significant investor interest from well-capitalized investors, but is still assessed to be less crowded than a public enterprise because of the relatively specialized nature of private markets.

Overall, Opportunity 3 is assessed to have the highest enterprise and investor contribution. Opportunity 2 is relatively scalable, but particularly crowded. Opportunity 1 is relatively neglected, but the least scalable due to the early stage, the natural scale of its product category and the exploratory nature of the investment.

Size matters

In the overall results table below, while all the impact returns and impact values are very different, the impact values are actually 'closer' than the returns. This is primarily because Opportunity 1 is much smaller in terms of the amount invested than Opportunity 2.

More scalable opportunities will tend to also be bigger investment opportunities. This means that, for equal impact returns, more scalable opportunities will offer more potential to generate impact value.

This is particularly important if there are fixed costs of investment. As an extreme example, suppose a private equity search fund only has capacity to make a single investment. Both because that is what it set out to do and because of the operational capacity of its team. If they could find an investment with the same impact return as Opportunity 2, but 100x larger (i.e. a \$10b investment), then this would be the best option to maximize impact value. This is because the scale of the investment trumps the much higher impact returns of the other opportunities.

In practice there will usually be a mix of fixed and variable costs, so both impact values and returns should be considered.

Impact Returns

The table below presents each key step and the resulting impact returns for the three cases. See the [appendix](#) for step-by-step details on each case.

Key Step		Opportunity 1 (Mosquito Nets)	Opportunity 2 (Education Technology)	Opportunity 3 (Covid Vaccine)
1	Potential Gross Impact	77k lives saved	260 lives saved	700k lives saved
	<i>Probability of success</i>	<i>95% for nets within loan; 20% outside of loan term</i>	89%	10%
	Expected Gross Impact	16.5k lives saved	230 lives saved	70k lives saved
	<i>Enterprise Size</i>	<i>\$1.625m</i>	<i>\$200m</i>	<i>\$70m</i>
	Expected Gross Impact per \$1m ²	10k lives saved	1.2 lives	1,000 lives
2	<i>Enterprise Contribution Multiplier</i>	10%	24%	74%
	Enterprise Impact per \$1m ²	1000 lives saved	0.3 lives saved	740 lives saved
	<i>Investor Contribution Multiplier</i>	11%	22%	47%
	Investor Impact per \$1m ³	110 lives saved	0.1 lives saved	330 lives saved
3	<i>Strategic Price of Impact</i>	\$5,000 per life saved		
	Impact Return	55%	0.03%	165%
	Impact Value	\$180k	\$30k	\$17m
	<i>Expected Holding Period</i>	5 years	5 years	5 years
	Impact Return (Annualized)	11%	0.01%	33%

² Per \$1m 'Enterprise Financial Size' as discussed in the main text.

³ Per \$1m invested.

Discussion

How should you interpret these impact returns? Opportunities should only be judged on the sum of their financial and impact returns. Depending on the financial returns, any of these opportunities could have the best overall 'impact-adjusted' return. We leave it as an 'exercise for the reader' to decide for yourself what you think the overall results are likely to be.

That said, assuming the SPI is set correctly, an opportunity with an impact return >100% is worth doing independent of the financial return (assuming it is at worst -100%). Beyond this, more work is needed to understand the distribution of impact returns across opportunities. However, based on our findings so far, and the fact that Acumen chose to highlight Opportunity 1 with a case study, we expect that opportunities with impact returns higher than this are exceptional.

These results highlight how impact returns can vary a lot across opportunities. Opportunity 1 has the highest Expected Gross Impact per dollar but this is offset by low expected contributions because of relatively weak scalability and crowdedness. In contrast, Opportunity 2 has a much lower Expected Gross Impact per dollar, but has reasonably good contribution multipliers because of the relative scalability of the growth equity stage. In the end it is Opportunity 3, which scores reasonably well on each dimension that has the highest expected impact return.

Opportunity 1 is a classic example from almost 20 years ago that we were able to analyze because of Acumen's pioneering leadership in sharing the case publicly. Acumen's original analysis, like ours, compared the impact of their investment to a relevant form of highly-effective charity (i.e. donating Malaria nets). They did not account for the 'contribution' dimension and as a result found the investment was some order of magnitude (~50x) more cost-effective than charity. In this light, we see our results as i) producing much more reasonable results, and ii) still finding that the investment was highly impactful and possibly better than a similar grant. If we had found impact returns of >100% per year, we would question both our own methodology and why Acumen did not invest even more in the opportunity.

Opportunity 2 is another classic example that we were able to analyze because of the TPG Rise team's pioneering leadership in sharing the case publicly. In their own analysis, TPG Rise used social costs, not opportunity costs, to inform their price of impact. This had the implicit effect of leaving them using the value of a statistical life in the US as the 'price of impact' - this is ~1000x higher than the price of impact we used in this report. Note that such a high 'price' may be a fine answer to the 'moral value' society should place on a life, but impact returns aren't about debating that. Rather impact returns are about the relatively objective question of how good an investment is given the availability of other opportunities for impact.

That said, we see TPG Rise's analysis, with its high price of impact and lack of adjustments for contribution, as underlining the strong case for the investment – just from a financial perspective not

an impact one. If the business is producing impacts that are highly valuable to US society (as determined by their social cost/value), then this gives it a strong license to operate and grow.

Additional important points of context

- **No second guessing.** We sought to use the original numbers from each case as much as possible —we did not re-evaluate the evidence for the gross impact in the original cases.
- **Uncertainty.** As with any model of the world, including qualitative ones, assessing impact returns involves making assumptions under great uncertainty. We aim to be transparent about our assumptions and uncertainties, as we believe that a transparent, speculative analysis is better than no analysis at all. As part of this, the models in the appendix use a tool called Guesstimate to run Monte Carlo simulations on the model parameters. This enables us to model each parameter with uncertainty and quickly visualize the effect on the end result. For simplicity, we only report average values in the table below.
- **Relative values.** The uncertainties involved mean one shouldn't look at these results as an attempt to work out the 'true' impact returns. Rather, the question is: was the assessment of each step consistent with how the other opportunities were analyzed? If so, then looking at the impact returns (plus expected financial returns) is a powerful way to assess the relative strength of each opportunity.
- **Additional considerations.** It can be useful to capture other considerations as returns. That is, to weigh other decision criteria in terms of how it would make sense to pay for it, given the effect it will have on your goals. This can include non-financial contributions and long-term strategic considerations. As these considerations depend greatly on context, we don't assess them in this report.
- **Direct vs Fund investments.** These cases have all been analyzed as direct investments. Investing in funds is a more common and practical choice for many asset owners. However, fund investments come with another layer of contribution to adjust for (i.e. "LP contribution"). We are happy to share examples of such LP impact return assessments on request.

We added Opportunity 3 to these classic examples both to be an example of an investment that 'scores well on each consideration' and to demonstrate that impact returns can be applied to high risk-high expected value opportunities. Sometimes we encounter views that quantitative approaches, like impact returns, are biased against such opportunities. We believe that nothing could be further from the truth. Rather such views themselves reflect a bias towards only using numbers for 'measurable' things rather than as tools for careful reasoning in general. Thus, this example serves to illustrate how one might apply impact returns to reason about such opportunities.

It is important to emphasize that the specific numbers are a result of our interpretation of each case. While we have aimed to be impartial in our interpretation, the actual investment teams involved in each case would have access to much more information. This could include information on non-financial investor impacts (e.g. engagement) or indirect impacts. Our goal here is to show one way that each case can be assessed, not to claim that our chosen numbers are *the* way.

Create your own version of these analyses using [this template](#):

- What '**contribution multipliers**' would you apply?
- What '**strategic price of impact**' would you use?

Appendix - Opportunity Walkthroughs

Step		Opportunity 1 (Mosquito Nets)	Opportunity 2 (Education Technology)	Opportunity 3 (Covid Vaccine)
Context		The opportunity was a 5-year \$325,000 loan at a 6% annual interest rate to A to Z Textile Mills in Tanzania. The loan was to enable A to Z to become the first African manufacturer of long-lasting insecticide-treated bednets (LLINs) that are more effective than traditional nets at protecting people against the mosquitoes that transmit malaria. Making the loan came with an estimated \$130,000 in disbursement and management costs, because of Acumen's hands-on support for their investments. The numbers and estimates are drawn from the public document found here.	EverFi is an education-technology company that was one of TPG Rise's first investments. For this case we use the representative data published by TPG Rise. Rise identified three EverFi programs that it focused on for its impact analysis. Among them, we focus for simplicity on an illustrative analysis of AlcoholEdu. AlcoholEdu is an online course designed to deter alcohol abuse among college students - TPG Rise estimated it to account for 28% of the expected impact of EverFi. Similar analysis could be applied to their sexual assault and financial literacy work. To translate the outcomes into dollar terms, TPG Rise turned to the U.S. Department of Transportation's guidance on valuing the reduction of fatalities or injuries, which uses a measure called the value of a statistical life. According to this guidance, a fatality is worth \$5.4 million. Thus, TPG Rise inferred that AlcoholEdu could expect to generate social value of at least \$194 million by saving 36 lives.	This example focuses on a venture that is working to rapidly develop a low-cost and easily-distributed vaccine for future variants of Covid-19. The success of this venture would result in i) producing a vaccine that works for the billions of people in low-income countries, and ii) proving that such a vaccine could be rapidly developed, paving the way for future vaccine developments to achieve this pace.
Model Link		Model spreadsheet with template		
1	Potential Gross Impact	First we estimate the number of LLINs A to Z could manufacture with a capital injection. This estimate is broken down into two time periods, during the loan and after the loan, to account for the future nets produced in the facility. The case study provided by Acumen estimates 2m LLINs during the loan. We then estimate 100m LLINs after the loan, as production scales up given success. This scenario accounts for the fact that the direct impact of the loan may be small compared to the scale of the market, but by making this first loan Acumen initiates a partnership with A to Z that can scale over time. To convert an 'output' of LLINs into Gross Impact we use GiveWell's cost-effectiveness analysis here of an expected outcome of 750 child's lives saved per 1m LLINs distributed. $(2m \times 750 \text{ expected outcome per } 1m) / 1m + (100m \text{ LLINs} \times 750 \text{ expected outcome per } 1m) / 1m = \mathbf{77k \text{ lives saved}}$	The original case estimates 2.175m students will use AlcoholEdu during the holding period (5 years) and assumes this continues for the following 5 years. This leads to a total estimate of ~4.4m students using the course. To convert this into impact, we estimate the resulting lives saved from using the course. We use the estimate of an 11% reduction in incidents per student. To convert this into lives saved we use the figure from the NIH indicating incidents of 0.015% resulting in death. Assuming AlcoholEdu's impact is the same percentage of EverFi's impact as in TPG Rise's estimate we then divide by the relative proportion (28%) to get an estimate of EverFi's overall gross impact. $(2.2m \text{ students} \times 2 \text{ holding periods}) \times 11\% \text{ reduction in incidents} \times 0.0152\% \text{ fatal incidents} / 28\% = \mathbf{260 \text{ lives saved}}$	The firm estimates that success of the venture will translate to 500m doses distributed. We take their lives saved per dose estimate of 0.0007 and apply a 2x heuristic multiplier to reflect its potential effect on future vaccines and distributions. We consider a simple success/fail pair of scenarios. $500m \times 0.0007 \times 2 = \mathbf{700k \text{ lives saved}}$

TOTAL PORTFOLIO PROJECT

Impact Returns Three Case Studies

	Expected Gross Impact	For the 2m LLINs produced during the loan an estimate of 95% is applied given the certainty of the technology. For the 100m LLINs produced after the loan a significantly lower probability of 20% is applied to account for the potential failure of the project and/or process afterwards. (95%*1,500 lives saved)+(20%*75k lives saved) = 16.5k lives saved	We apply an 89% probability of success, in line with the original case. 89% * 260 lives saved = 230 lives saved	We apply a 10% success probability based on biotech industry rates (note the firm's internal estimate would, naturally, be higher). 700k lives saved * 10% probability of success = 70k lives saved
	Expected Gross Impact per dollar	We divide the Expected Gross Impact by our estimated valuation of the venture, \$1.625m. 16.5k lives saved/\$1.625m = 10k lives saved per \$1m	We divide the Expected Gross Impact by the expected valuation of the venture, \$200m. 230 lives saved/\$200m = 1.2 lives saved per \$1m	We divide the EGI by the expected valuation of the venture, \$70m (post-money). 70k lives saved/ \$70m = 1,100 lives saved per \$1m
2	Contribution	We now adjust for enterprise and investor contribution multipliers. Values for the contribution scores can be found in the corresponding section .		
	<i>Enterprise Contribution Multiplier</i>	10%. LLINs were a new technology at the time. So, while Chinese and Vietnamese competitors were active, LLIN production was still relatively neglected but also not necessarily that scalable (even if there was plenty of room for scale, in the future, once demand for LLINs grew).	24%. A higher than average scalability given the company's services are largely online, it is growing with a large addressable market, and could plausibly grow faster with more investment. Only slightly below average crowdedness as while the firm's growth indicates the unique value of its product, there is clear potential for viable alternatives in online and offline courses.	74%. The category of rapid coronavirus vaccine development for low-income countries seems to be less crowded than the venture itself, but it does attract a wider group of funders including governments. The category itself is naturally more scalable than any single enterprise - the world likely would benefit from even more ventures like this trying different approaches.
	<i>Investor Contribution Multiplier</i>	11%. The business is fundamentally less crowded than a US public company as it is in EM and private. It is entering a scalable space, but it is not clear that it needs to dramatically scale immediately. Rather it needs just a few machines to get going and establish product-market fit (remember there is a difference between an enterprise's scalability and the scale of its potential market). The revelation in future years that the procurement rules of major buyers were biased against local African manufacturers underlines how more immediate scale wouldn't necessarily have produced much more impact.	22%. Above average scalability as a growth enterprise. Lower crowdedness than for a public company, but not by much given reports TPG Rise had to work hard to secure the deal.	47%. This opportunity is relatively less crowded as only a few investors with biotech expertise are interested in taking the risk. But less scalable given an estimate that increasing its scale by \$20m would cut its return roughly in half.
	Investor Impact per \$1m	36k lives saved per \$1m * 9% * 11% = 110 lives saved per \$1m	2.3 lives saved per \$1m * 24% * 22% = 0.06 lives saved per \$1m	1,100 lives saved per \$1m * 71% * 47% = 330 lives saved per \$1m
3	Strategic Pricing	We use \$5,000 per life as the Strategic Price of Impact, in line with the published policy of Open Philanthropy, which was informed by GiveWell analyses.		
	Impact Value & Impact Return (Cumulative)	110 lives saved per \$1m * \$5000 per life = 55% We take the amount invested in the opportunity, \$325k, and multiply it by the impact per dollar. \$325,000 investment amount * 55% = \$180,000.	0.06 lives saved per \$1m * \$5000 per life = 0.03% We take the amount invested in the opportunity, \$100m, and multiply it by the impact per dollar. \$100m * 0.03% = \$30,000.	330 lives saved per \$1m * \$5000 per life = 165% We take the amount invested in the opportunity, \$10m, and multiply it by the impact per dollar. \$10m * 165% = \$16.5m.
	<i>Expected Holding Period</i>	5 years		
	Impact Return (Annualized)	Annualized: 55% / 5 years = 11%	Annualized: 0.03% / 5 years = 0.01%	Annualized: 165% / 5 years = 33%